BEATRICE LOWMAN

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SKILLS

Neurotechnology & Signal Processing: BCI, EEG/iEEG decoding, neuroimaging ML analysis, facial landmark extraction, GSR, Tobii eye-tracking
ML & Data: Python, Java, SQL, MQL, deep learning, computer vision, transformers, LLMs, real-time pipelines, data engineering, visualization
Clinical Research: IRB protocols, CITI certified (Human Subjects), REDCap, BioLAB, PsychoPy, Inquisit, E-Prime, cognitive task administration

EDUCATION

University of California, Berkeley **May 2026**
B.A. in Cognitive Science and B.A. in Data Science
Relevant Coursework: Deep Learning; Natural Language Processing; Data Engineering; Computational Models of Cognition; Cognitive Neuroscience; Mind, Brain, and Identity; Human Contexts and Ethics of Data

PROJECTS

Neurotech@Berkeley | Software Lead

Early Dementia Detection via Hybrid CNN-ViT Model **Spring 2026**
- Trained MedViT-2 — a hybrid CNN-Transformer architecture — for four-class dementia stage classification from neuroimaging data, the first evaluation of MedViT on neuroimaging tasks
- Model achieved F1 scores >99% across all classes with overall accuracy >99%, outperformed existing baselines in early dementia stages; results presented at C.T.N.C. 2026

Brain-to-Text: Neural Speech Decoding from iEEG **Fall 2025**
- Built on the Brain-to-Text Benchmark '24 to decode intracranial EEG signals from an ALS patient into intended sentences using an RNN-GRU + N-Gram + LLM rescoring pipeline
- Engineered optimizations including temporal masking, diphone lexicon, rebuilt CTC decoding, RNNT loss, and Cold Fusion LLM integrating phoneme embeddings with LLaMA 3.3 hidden states

3D MRI Brain Tumor Segmentation **Fall 2024 – Spring 2025**
- Developed a U-Net FCN to segment brain 3D MRI scans from the BraTS dataset; iterated from standard U-Net with Dice Loss to attention-gated U-Net with Focal Tversky Loss
- Identified critical regions including enhancing tumor, necrotic tumor core, and peritumoral edema with significantly above-chance accuracy

Publications Member **2023**
- Published in MIND Magazine and featured on the Neurotalk Spotify Podcast on EEG-guided meditation and AI-generated artwork

EXPERIENCE

Google DeepMind AI UX Team **Sept 2025 – March 2026**
Project Manager & AI Systems Engineer — Contract *Berkeley, CA*
- Designed a trust-adaptive AI framework to dynamically optimize agent autonomy for user comfort with contextual bandit and MDP models
- Developed a natural language Google Calendar AI agent for intelligent scheduling inference, conflict resolution, and user support
- Architected a modular system decoupling agent capabilities from trust-based control logic, enabling reuse across broader AI systems

Xiberlinc Inc. **Jan 2025 – Aug 2025**
Neural Engineering Intern *Remote*
- Implemented back-end software for a novel non-invasive BCI prototype designed to enhance emotional wellbeing and cognition
- Developed a real-time pipeline for video-based biosignal feature extraction, processing, and analysis using MediaPipe and OpenCV
- Engineered interpretable features via PCA, ICA, and action coding; trained LSTM and SVM models on EEG data to predict neural states

A.C.C. Senior Services **2020; 2025 – 2026**
Data Engineering Intern *Sacramento, CA*
- 2025-2026: Streamlined and automated volunteer / participant matching systems; re-designed database to improve operational efficiency
- 2020: Used a data-driven approach to identify and head inclusion initiatives for undercounted populations and zones in the U.S. Census

RESEARCH EXPERIENCE

UC Berkeley CALM Lab **Sept 2024 – June 2025**
Research Assistant & Computer Programming Specialist *Berkeley, CA*
- Contributed to the N-ACT randomized controlled trial — a cognitive remediation intervention for emotion-related impulsivity — under Dr. Sheri Johnson and Dr. JD Allen
- Developed digital training tasks including the adaptive Emotional Stop-Signal Task (ESST) and additional hot/cold cognition paradigms
- Conducted participant sessions under IRB protocol: cognitive tasks, GSR and eye-tracking data collection, and self-report measures
- Developed and presented materials showcasing experimental design and progress at the Bay Area Affective Sciences conference (2025)

SELECTED PRESENTATIONS

Lowman, B., et al. Improving Dementia Stage Classification with Hybrid CNN-ViT Model. Poster, C.T.N.C., 2026.
Lowman, B., et al. Computer Vision for Brain Tumor Detection. Poster, C.T.N.C., 2025.
Marquez-Garcia, M., ..., Lowman, B., et al. Hot Thoughts, Smart Moves: Neurobehavioral Training for Rumination and Impulsivity. Poster, B.A.A.S., 2025.